SignBridge — how it works

One laptop · one screen facing the Deaf guest · the receptionist only talks and listens

■ Intent classification (highlighted)

The receptionist speaks → the guest sees ASL

1 Speak
The receptionist holds Space and talks
Microphone

2 Transcribe
The speech becomes text
OpenAI Whisper

3 Understand
What does the receptionist mean?
Intent classifier

4 Translate
The text becomes ASL signs
ASL gloss

5 Sign
A 3D bellhop signs it, with captions
3D avatar

The guest signs → the receptionist hears English

1 Sign
The guest signs to the webcam
Camera

2 Track
Hands, body and face are tracked
MediaPipe

3 Recognize
Each sign is recognized
Our sign classifier

4 Understand
Which sentence does the guest mean?
Intent decoder

5 Speak
The answer is said out loud
ElevenLabs voice

87
signs the avatar can play, recorded by the team

1,320
training takes recorded by 5 signers

≈ 90 %
of sentences understood for a signer never seen

SignBridge — technical workflow

One laptop · one screen facing the Deaf guest · the receptionist only talks and listens

■ Intent classification (highlighted)

RECEPTIONIST → GUEST speech to ASL

1 Push-to-talk

Space held = record

- MacBook built-in mic, chosen automatically (AirPods / iPhone avoided)
- MediaRecorder → webm audio
- the room noise is never transcribed

2 Speech to text

OpenAI Whisper (whisper-1)

- Next.js route /api/transcribe, key server-side
- fallback: Chrome Web Speech API

3 INTENT CLASSIFIER

20 receptionist intents (hotel front desk)

- keyword patterns + resemblance to the reference sentence (≥ 1/3 of its content words)
- slots: room number, times
- web/lib/voice/intents.ts · /api/gloss → { gloss, intents }

4 Text → ASL gloss

intent → curated ASL gloss

- ASL word order, BROWS-UP / BROWS-DOWN on questions, numbers
- unknown sentence → word by word: 87 known signs, synonyms, plurals, else fingerspelling (FS:WORD)

5 3D signing avatar

Three.js · rigged bellhop (GLB)

- Tripo mesh, Mixamo skeleton, 3 phalanges per finger
- replays 87 signs the team recorded with MediaPipe
- retargeting: arm IK, wrist basis, phalanx aiming · transitions · 30 fps · karaoke captions

GUEST → RECEPTIONIST ASL to speech

1 Guest turn & capture

recognition/live_key.py · OpenCV

- starts when the avatar has finished signing
- webcam 1280×720; ends when the hands rest 2.5 s
- live MJPEG preview on the page (green frame)

2 Landmarks

MediaPipe Holistic 1.1.0, on-device

- 33 pose + 2 × 21 hand + 26 face points
- normalized on the shoulders, 30 fps, cleaned
- no video leaves the laptop

3 Sign classifier

PyTorch, 22 classes (21 signs + NONE)

- trained on 1,320 takes by 5 signers (12 per sign each)
- 141 features per frame, resampled to 32 steps
- scores sliding 1 s windows (step 0.13 s)

4 INTENT DECODER

22 guest sentences = guest intents

- aligns each candidate sentence on the window probabilities (monotonic dynamic programming)
- the sentence context fixes misread signs
- recognition/sentence.py

5 Text to speech

ElevenLabs (eleven_flash_v2_5)

- Next.js route /api/tts
- fallback: browser speechSynthesis
- spoken on the laptop + “You said: …” on screen

Turn-taking web/lib/turns.ts

- FRONT DESK → REQUESTING → GUEST → SPEAKING → FRONT DESK
- microphone muted for the whole guest turn and while the answer is spoken
- recognizer events: live_key → POST /api/guest/signs → SSE /api/guest/stream → page
- timeouts: recognizer silent 3 s, guest turn 30 s

Stack

- Web: Next.js 16 · React 19 · TypeScript · Three.js
- Recognition: Python 3.12 · MediaPipe 1.1.0 · OpenCV · PyTorch
- Cloud: OpenAI Whisper (speech) · ElevenLabs (voice)
- Data: 87 avatar signs + 1,320 training takes recorded by the team

Results

- Isolated signs, unseen signer (leave-one-out): 89.8 % right · top-3 97.1 %
- Continuous sentences, unseen signer (simulated): ≈ 90 %
- Live rehearsal, no guidance: 8 / 8 guest sentences
- Decoding < 0.1 s after the hands rest; voice ≈ 0.5 s